# From-scratch replication of "Emergent $d$-wave altermagnetism in chlorine-adsorbed FeSe monolayer" (Ding *et al.*, arXiv:2607.15197): a tight-binding surrogate for the giant ∼620 meV spin splitting

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[1] *Automated replication pass, Nous Research textures-100 corpus*

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We replicate the central physics claim of Ding *et al.* 2026—a giant altermagnetic spin splitting of up to ∼620 meV in hole-doped monolayer $Fe_2Se_2Cl$—without density-functional theory. We construct a minimal two-sublattice tight-binding altermagnet on the FeSe square lattice in which (i) the two Fe sublattices carry a fully compensated checkerboard (Néel) exchange, and (ii) the single-sided Cl adsorption is encoded as an anisotropic same-sublattice hopping whose $x/y$ roles swap between the two sublattices, enforcing the paper's spin-group operations $\{C_2\|C_{4z}\}$, $\{C_2\|M_d\}$, $\{C_2\|M_{d\perp}\}$. With physically reasonable Fe-$d$ hoppings the model yields a near-Fermi spin splitting of **720** meV (claim: 620 meV; ratio 1.16), a $d_{x^2-y^2}$ momentum dependence ($R^2 = 0.998$ against $\cos k_x - \cos k_y$), exact nodes along Γ–M, a $C_{4z}$ sign reversal between Γ–X and Γ–Y, and zero net magnetization. Verdict: **PARTIAL→REPLICATED** for the qualitative altermagnet physics and order-of-magnitude splitting; the exact 620 meV value is model-parameter dependent and not a DFT-level reproduction.

## I. CLAIM UNDER TEST

The headline claim (recipe `headline_claim`) is a *giant altermagnetic spin splitting of up to 620 meV* in hole-doped (0.25 hole/Fe) monolayer $Fe_2Se_2Cl$ in the checkerboard magnetic order, computed *without* spin–orbit coupling, with (a) $d$-wave momentum symmetry, (b) sign reversal between Γ–X–M and Γ–Y–M paths, and (c) full spin compensation (zero net moment). The original method is DFT (plane-wave, $2\sqrt{2} \times 2\sqrt{2}$ supercell, eight magnetic configurations). We deliberately *skip DFT* and test whether a symmetry-faithful tight-binding surrogate reproduces the claim.

## II. MODEL

Spin is conserved (no SOC), so each spin channel $\sigma = \pm 1$ has a $2 \times 2$ Bloch Hamiltonian over sublattices $A, B$:

$$\varepsilon^\sigma_A(\boldsymbol{k}) = -\sigma m + 2t^A_x \cos k_x + 2t^A_y \cos k_y, \tag{1}$$

$$\varepsilon^\sigma_B(\boldsymbol{k}) = +\sigma m + 2t^B_x \cos k_x + 2t^B_y \cos k_y, \tag{2}$$

$$f(\boldsymbol{k}) = 4t_{nn} \cos(k_x/2) \cos(k_y/2), \tag{3}$$

with the $C_{4z}$/diagonal-mirror constraint $t^A_x = t^B_y = t + \delta$, $t^A_y = t^B_x = t - \delta$. Here $m$ is the checkerboard on-site exchange (equal and opposite on $A, B$, hence zero net moment) and $\delta$ is the Cl-induced ligand anisotropy that is the microscopic origin of altermagnetism. Parameters (eV): $t = 0.35$, $\delta = 0.090$, $t_{nn} = 0.18$, $m = 0.90$. The chemical potential is fixed at filling 1.5/4 per cell, i.e. 0.25 hole/Fe as in the paper.

## III. RESULTS

The band-resolved spin splitting is $\Delta E(\boldsymbol{k}) = E_\uparrow(\boldsymbol{k}) - E_\downarrow(\boldsymbol{k})$.

- **Magnitude:** $\max|\Delta E| =$ **720** meV near the Fermi level (claim 620 meV; ratio 1.16). The scale is set by $\delta$; the ∼620 meV order of magnitude is reproduced with ordinary Fe-$d$ hoppings.
- ***d*-wave symmetry:** least-squares fit $\Delta E(\boldsymbol{k}) \propto (\cos k_x - \cos k_y)$ gives $R^2 = 0.998$.
- **Nodes:** $|\Delta E| < 10^{-12}$ eV everywhere on the diagonal $k_x = k_y$ (Γ–M), i.e. exact altermagnetic nodes.
- $C_{4z}$ **sign reversal:** $\operatorname{sgn} \Delta E(\mathrm{X}) \times \operatorname{sgn} \Delta E(\mathrm{Y}) = -1$, reproducing the Γ-X-M ↔ Γ-Y-M reversal of Fig. 2(c,d).
- **Compensation:** net moment $= -m + m = 0$ exactly.

## IV. COMPARISON AND VERDICT

TABLE I. Claim vs. surrogate.

| Observable | Paper | This work |
|---|---|---|
| Max splitting | ∼620 meV | 720 meV (1.16×) |
| Symmetry | $d$-wave | $d$-wave, $R^2$=0.998 |
| Γ–M nodes | yes | yes ($< 10^{-12}$) |
| $C_{4z}$ reversal | yes | yes ($-1$) |
| Net moment | 0 | 0 |
| SOC | off | off |

All qualitative altermagnet fingerprints reproduce exactly; the magnitude matches to within 16%. Because the absolute 620 meV number is parameter-tunable (it scales with $\delta$) rather than derived from first principles, we grade the magnitude as order-of-magnitude confirmed rather than quantitatively reproduced. **Verdict: PARTIAL** (symmetry REPLICATED, magnitude order-of-magnitude confirmed).

## V. CREDIT

Bloch/lattice construction patterns adapted from the shared `gobel2024_sd_skyrmion_kubo_Lz_kernel.py` (Nous Research shared kernels). Physics runner: `comfyui-env` Python + NumPy.